

1

Programs

Addition

```

P add: ( $x_{ip+2} + x_{ip+3}$ )
   $x_{ip+1} := x_{ip+2};$ 
  loop  $x_{ip+3}$  do
     $x_{ip+1} := \text{suc}(x_{ip+1})$ 
  end
  // cleanup

```

Multiplication

```

P mult: ( $x_{ip+2} \cdot x_{ip+3}$ )
   $x_{ip+1} := 0;$ 
  loop  $x_{ip+3}$  do
     $x_{ip+1} := x_{ip+1} + x_{ip+2}$ 
  end
  // cleanup

```

Exponentiation

```

P exp: ( $x_{ip+2}^{x_{ip+3}}$ )
   $x_{ip+1} := 0;$ 
   $x_{ip+1} := \text{suc}(x_{ip+1});$ 
  loop  $x_{ip+3}$  do
     $x_{ip+1} := x_{ip+1} \cdot x_{ip+2}$ 
  end
  // cleanup

```

Evaluation Traces

Addition

$$\begin{aligned}
 & \delta((0, 3, 2, 0, \dots), \text{P add}) = \\
 & \delta((0, 3, 2, 0, \dots), x_{ip+1} := x_{ip+2}; \text{loop } x_{ip+3} \text{ do } x_{ip+1} := \text{suc}(x_{ip+1}) \text{ end}) = \\
 & \delta(\delta((0, 3, 2, 0, \dots), x_{ip+1} := x_{ip+2}), \text{loop } x_{ip+3} \text{ do } x_{ip+1} := \text{suc}(x_{ip+1}) \text{ end}) = \\
 & \delta((3, 3, 2, 0, \dots), \text{loop } x_{ip+3} \text{ do } x_{ip+1} := \text{suc}(x_{ip+1}) \text{ end}) = \\
 & \delta((3, 3, 2, 0, \dots), x_{ip+1} := \text{suc}(x_{ip+1}); x_{ip+1} := \text{suc}(x_{ip+1})) = \\
 & \delta(\delta((3, 3, 2, 0, \dots), x_{ip+1} := \text{suc}(x_{ip+1})), x_{ip+1} := \text{suc}(x_{ip+1})) = \\
 & \delta((4, 3, 2, 0, \dots), x_{ip+1} := \text{suc}(x_{ip+1})) = \\
 & (5, 3, 2, 0, \dots)
 \end{aligned}$$

Multiplication

$$\begin{aligned}
& \delta((0, 3, 2, 0, \dots), \text{P mult}) = \\
& \delta((0, 3, 2, 0, \dots), x_{ip+1} := 0; \text{loop } x_{ip+3} \text{ do } x_{ip+1} := x_{ip+1} + x_{ip+2} \text{ end}) = \\
& \delta(\delta((0, 3, 2, 0, \dots), x_{ip+1} := 0), \text{loop } x_{ip+3} \text{ do } x_{ip+1} := x_{ip+1} + x_{ip+2} \text{ end}) = \\
& \delta((0, 3, 2, 0, \dots), \text{loop } x_{ip+3} \text{ do } x_{ip+1} := x_{ip+1} + x_{ip+2} \text{ end}) = \\
& \delta((0, 3, 2, 0, \dots), x_{ip+1} := x_{ip+1} + x_{ip+2}; x_{ip+1} := x_{ip+1} + x_{ip+2}) = \\
& \delta(\delta((0, 3, 2, 0, \dots), x_{ip+1} := x_{ip+1} + x_{ip+2}), x_{ip+1} := x_{ip+1} + x_{ip+2}) = \\
& \delta((3, 3, 2, 0, \dots), x_{ip+1} := x_{ip+1} + x_{ip+2}) = \\
& (6, 3, 2, 0, \dots)
\end{aligned}$$

Exp

$$\begin{aligned}
& \delta((0, 2, 3, 0, \dots), \text{P exp}) = \\
& \delta((0, 2, 3, 0, \dots), x_{ip+1} := 0; x_{ip+1} := \text{suc}(x_{ip+1}); \text{loop } x_{ip+3} \text{ do } x_{ip+1} := x_{ip+1} \cdot x_{ip+2} \text{ end}) = \\
& \delta(\delta((0, 2, 3, 0, \dots), x_{ip+1} := 0), x_{ip+1} := \text{suc}(x_{ip+1}); \text{loop } x_{ip+3} \text{ do } x_{ip+1} := x_{ip+1} \cdot x_{ip+2} \text{ end}) = \\
& \delta((0, 2, 3, 0, \dots), x_{ip+1} := \text{suc}(x_{ip+1}); \text{loop } x_{ip+3} \text{ do } x_{ip+1} := x_{ip+1} \cdot x_{ip+2} \text{ end}) = \\
& \delta(\delta((0, 2, 3, 0, \dots), x_{ip+1} := \text{suc}(x_{ip+1})), \text{loop } x_{ip+3} \text{ do } x_{ip+1} := x_{ip+1} \cdot x_{ip+2} \text{ end}) = \\
& \delta((1, 2, 3, 0, \dots), \text{loop } x_{ip+3} \text{ do } x_{ip+1} := x_{ip+1} \cdot x_{ip+2} \text{ end}) = \\
& \delta((1, 2, 3, 0, \dots), x_{ip+1} := x_{ip+1} \cdot x_{ip+2}; x_{ip+1} := x_{ip+1} \cdot x_{ip+2}; x_{ip+1} := x_{ip+1} \cdot x_{ip+2}) = \\
& \delta(\delta((1, 2, 3, 0, \dots), x_{ip+1} := x_{ip+1} \cdot x_{ip+2}), x_{ip+1} := x_{ip+1} \cdot x_{ip+2}; x_{ip+1} := x_{ip+1} \cdot x_{ip+2}) = \\
& \delta((2, 2, 3, 0, \dots), x_{ip+1} := x_{ip+1} \cdot x_{ip+2}; x_{ip+1} := x_{ip+1} \cdot x_{ip+2}) = \\
& \delta(\delta((2, 2, 3, 0, \dots), x_{ip+1} := x_{ip+1} \cdot x_{ip+2}), x_{ip+1} := x_{ip+1} \cdot x_{ip+2}) = \\
& \delta((4, 2, 3, 0, \dots), x_{ip+1} := x_{ip+1} \cdot x_{ip+2}) = \\
& (8, 2, 3, 0, \dots)
\end{aligned}$$

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1.

P add: ($x_{ip+2} + x_{ip+3}$)

$M_1 :$ $x_{ip+1} := x_{ip+2};$

$M_2 :$ **if** x_{ip+3} **goto** $M_4;$

$M_3 :$ **halt;**

$M_4 :$ $x_{ip+1} := \text{suc}(x_{ip+1});$

$M_5 :$ $x_{ip+3} := \text{pred}(x_{ip+3});$

$M_6 :$ **if** x_{ip+1} **goto** $M_2;$

P fib: (x_{ip+2} . Fibonacci Zahl)

$M_1 :$ $x_{ip+1} := \text{suc}(0);$

$M_2 :$ $x_{ip+3} := \text{suc}(0);$

$M_3 :$ $x_{ip+2} := \text{pred}(\text{pred}(x_{ip+2}));$

$M_4 :$ **if** x_{ip+2} **goto** $M_6;$

$M_5 :$ **halt;**

$M_6 :$ $x_{ip+4} := x_{ip+1};$

$M_7 :$ $x_{ip+1} := x_{ip+1} + x_{ip+3};$

$M_8 :$ $x_{ip+3} := x_{ip+4};$

$M_9 :$ $x_{ip+2} := \text{pred}(x_{ip+2});$

$M_{10} :$ **if** x_{ip+1} **goto** $M_4;$

2.

P mult: $(x_{ip+2} \cdot x_{ip+3})$

$M_1 :$ $x_{ip+1} := x_{ip+2};$

$M_2 :$ **if** x_{ip+3} **goto** $M_4;$

$M_3 :$ **halt;**

$M_4 :$ $x_{ip+1} := x_{ip+1} + x_{ip+3};$

$M_5 :$ $x_{ip+3} := \text{pred}(x_{ip+3});$

$M_6 :$ **if** x_{ip+1} **goto** $M_2;$

P fac: $(x_{ip+2}!)$

$M_1 :$ $x_{ip+1} := \text{suc}(0);$

$M_3 :$ **if** $\text{pred}(x_{ip+2})$ **goto** $M_5;$

$M_4 :$ **if** x_{ip+1} **goto** $M_{10};$

$M_5 :$ **if** x_{ip+2} **goto** $M_7;$

$M_6 :$ **halt;**

$M_7 :$ $x_{ip+1} := x_{ip+1} \cdot x_{ip+2};$

$M_8 :$ $x_{ip+2} := \text{pred}(x_{ip+2});$

$M_9 :$ **if** x_{ip+1} **goto** $M_5;$

$M_{10} :$ **if** x_{ip+1} **goto** $M_{10};$

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Fibonacci

Das GOTO-Programm

P_{fib} : $(x_{ip+2} \cdot \text{Fibonacci-Zahl})$

$P_{\text{setup}};$

if x_{ip+2} **goto** $\dots;$

$P_{\text{rep. add}};$

$x_{ip+2} := \text{pred}(x_{ip+2});$

if x_{ip+1} **goto** $\dots;$

lässt sich durch das folgende WHILE-Programm darstellen:

P_{fib} : $(x_{ip+2} \cdot \text{Fibonacci-Zahl})$

$P_{\text{setup}};$

while $x_{ip+2} \neq 0$ **do**

$P_{\text{rep. add}};$

$x_{ip+2} := \text{pred}(x_{ip+2});$

end

Die einzelnen Programmblöcke bleiben unverändert. Die Äquivalenz beider Programme ergibt sich daraus, dass die Schleifenbedingung direkt übernommen wird und die Modifikation der Schleifenvariablen in beiden Fällen identisch erfolgt.

Factorial

```
P fac: ( $x_{ip+2}!$ )  
   $x_{ip+1} := \text{suc}(0);$   
  while  $\text{pred}(x_{ip+2}) == 0$  do  
  end  
  while  $x_{ip+2}$  do  
     $x_{ip+1} := x_{ip+1} \cdot x_{ip+2};$   
     $x_{ip+2} := \text{pred}(x_{ip+2});$   
  end
```

Die Fakultäts-Funktion wurde auf die selbe Version konvertiert. Die rechnenden Code Blöcke sind identisch geblieben, sowie die Terminations Condition.